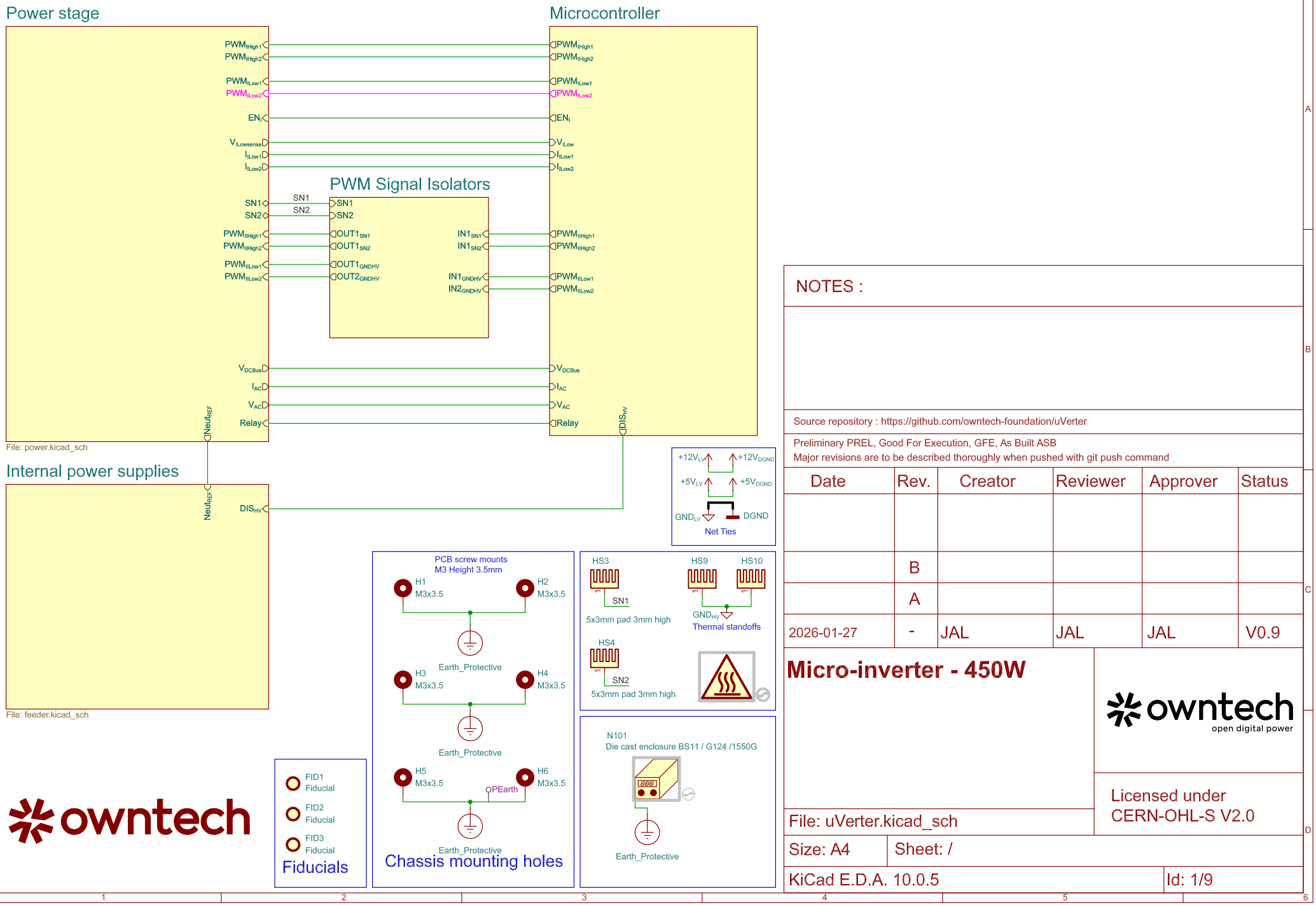


NOTES :

Source repository : https://github.com/owntech-foundation/uVerter

Preliminary PREL, Good For Execution, GFE, As Built ASB
Major revisions are to be described thoroughly when pushed with git push command

Date	Rev.	Creator	Reviewer	Approver	Status
	B				
	A				
2026-01-27	-	JAL	JAL	JAL	V0.9

Micro-inverter - 450W

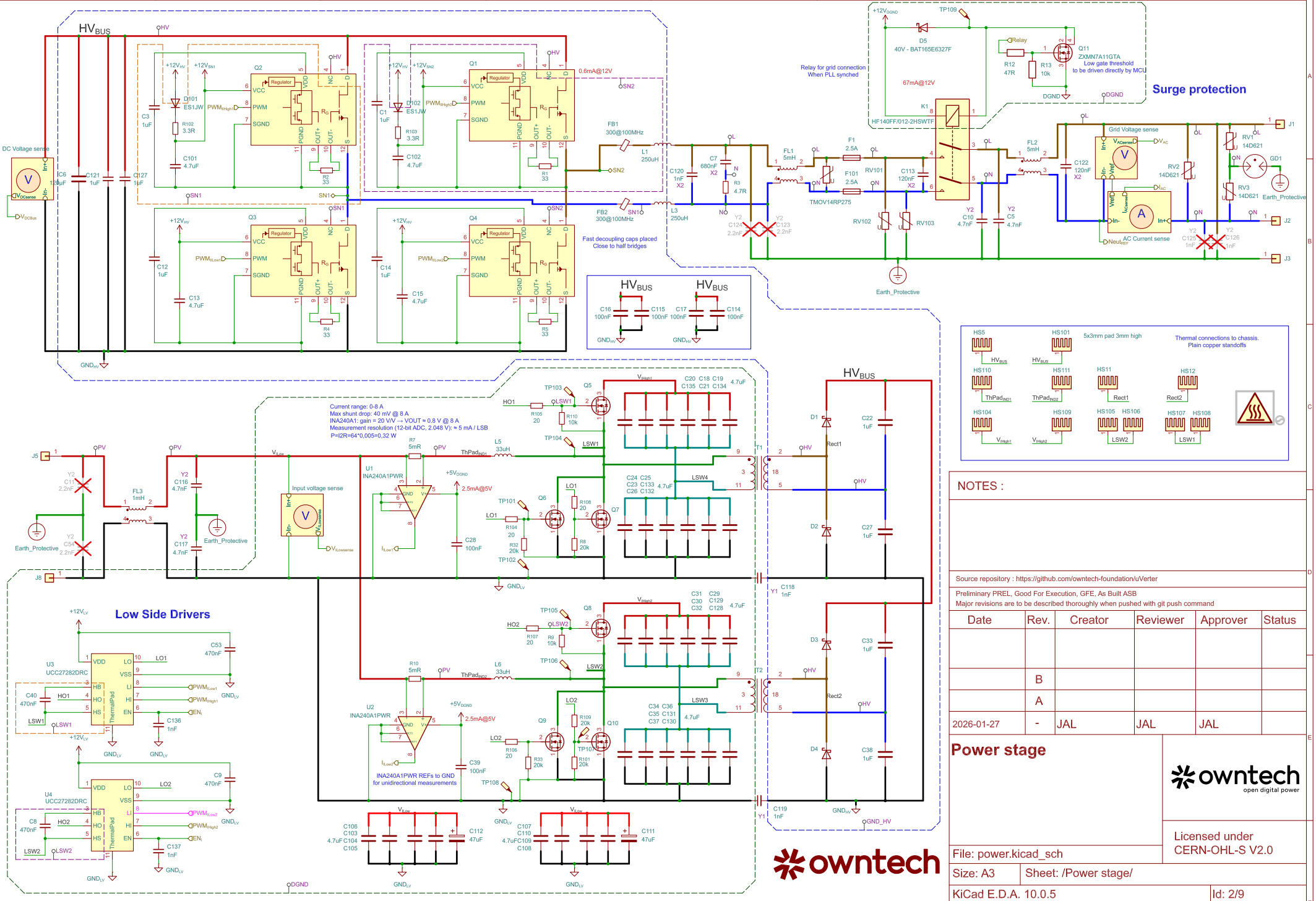
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File: uVerter.kicad_sch

Size: A4 | Sheet: /

KiCad E.D.A. 10.0.5 | Id: 1/9



NOTES :

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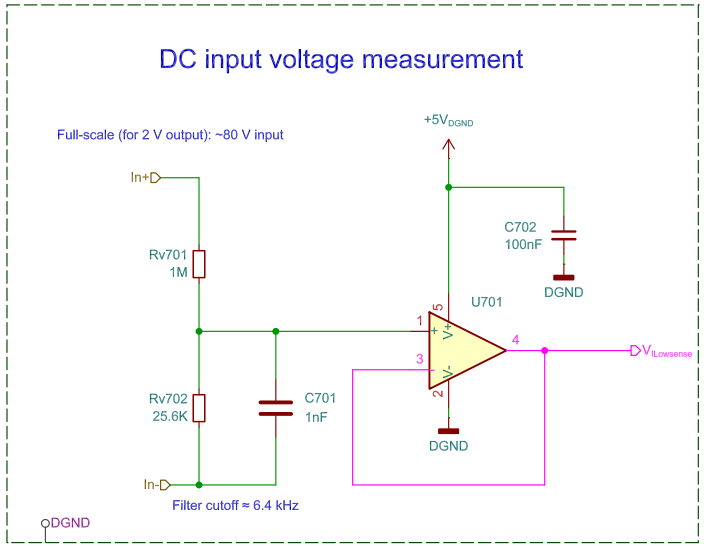
Power stage

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File: power.kicad_sch

Size: A3 | Sheet: /Power stage/

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NOTES :

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2026-01-27	-	JAL	JAL	JAL	

DC Input voltage measurement

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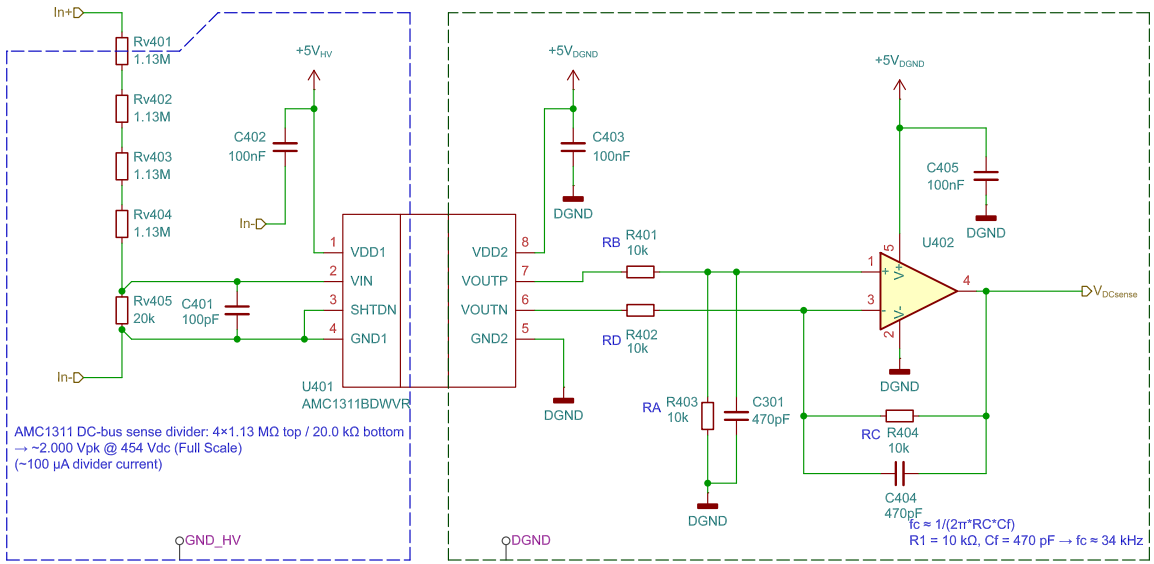
File: input_voltage_sense.kicad_sch

Size: A4 | Sheet: /Power stage/Input voltage sense/

KiCad E.D.A. 10.0.5 | Id: 3/9

DC Bus Voltage measurement

Simulation available
https://tinyurl.com/yodepbhp



Circuit can be seen as non inverting opamp
with gain : G = 1+(RC/RD) and a potential on non-inverting
input equal to :

V+ =(VOUTP-VOUTN)*(RA/(RA+RB))

V_{Diff} = (VOUTP-VOUTN)

V+ = V_{Diff}*(RA/(RA+RB))
V+ = V_{Diff}*(27k/(27k+27k))
V+ = V_{Diff}*0.5

NOTES :

Source repository : https://github.com/owntech-foundation/uVerter

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	A				
2026-01-27	-	JAL	JAL	JAL	

DC Bus voltage measurement

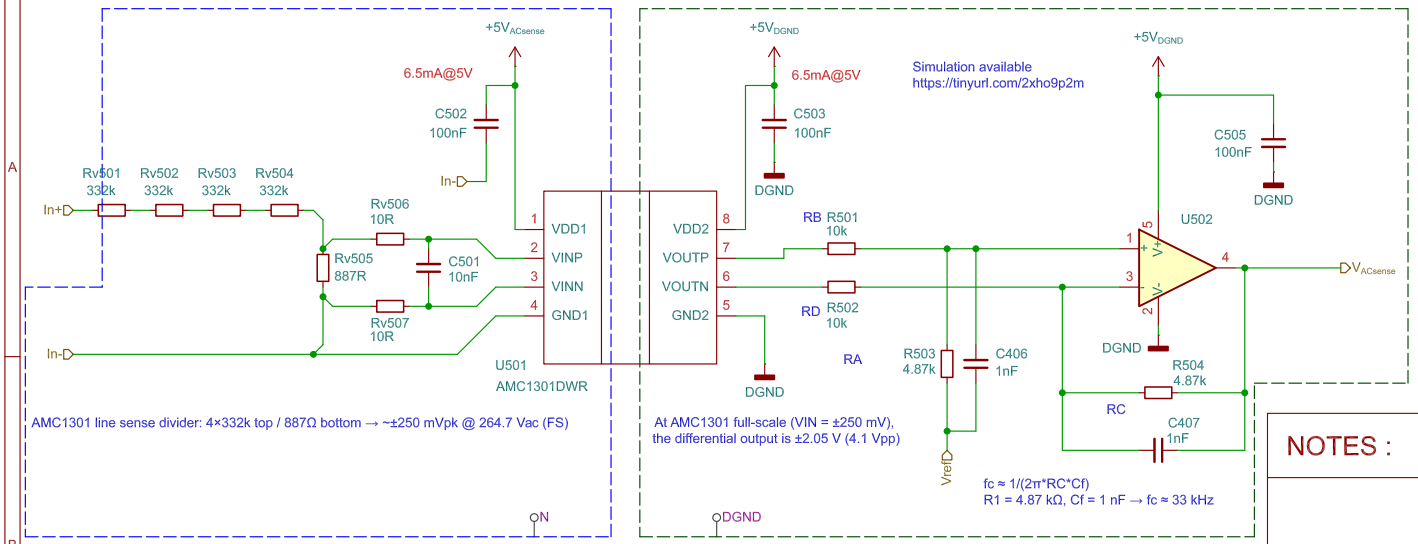
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File: DC_HV_voltage_sense.kicad_sch

Size: A4 | Sheet: /Power stage/DC Voltage sense/

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Inverter AC voltage measurement

VOUT=((RB/RC)VOUTP−(RA/RD)VOUTN)+VREF

If RB=RC and RA=RD
VOUT=G(VOUTP−VOUTN)+VRef
With VRef = 1V
G= 4.87k/10k = 0.487

VOUT = 0.487(VINP-VINN)+1V
output ≈ 0.00-2.00 V at AMC1301 full-scale

NOTES :

Source repository : https://github.com/owntech-foundation/uVerter

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Date	Rev.	Creator	Reviewer	Approver	Status
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	A				
2026-01-27	-	JAL	JAL	JAL	

Grid voltage measurement

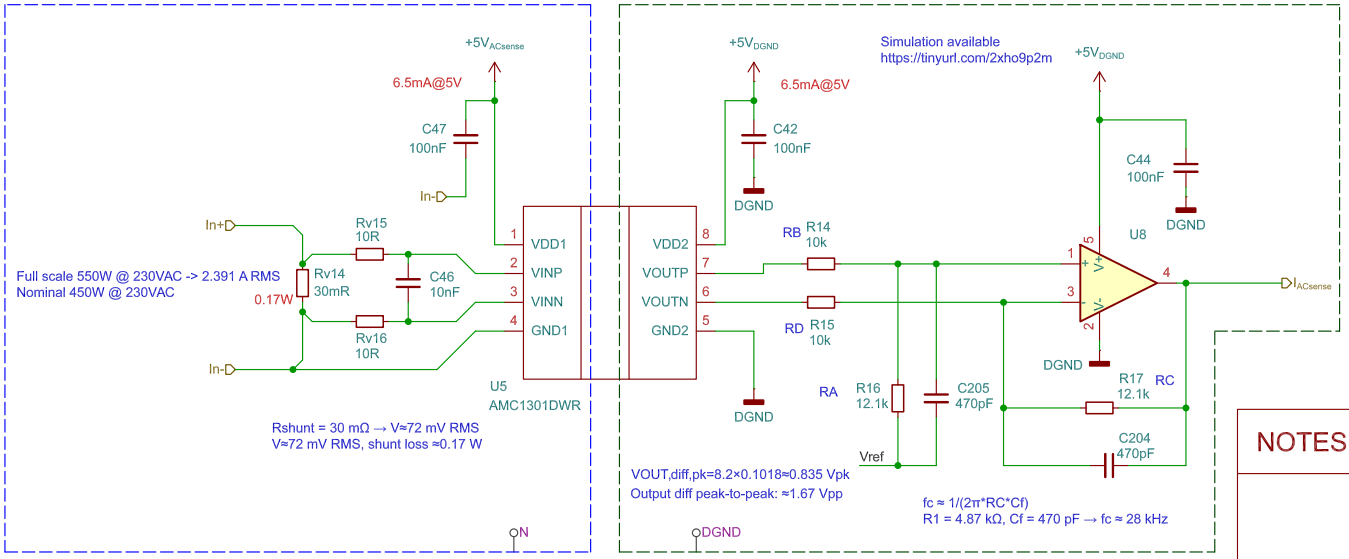
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File: AC_voltage_sense.kicad_sch

Size: A4 | Sheet: /Power stage/Grid Voltage sense/

KiCad E.D.A. 10.0.5 | Id: 5/9

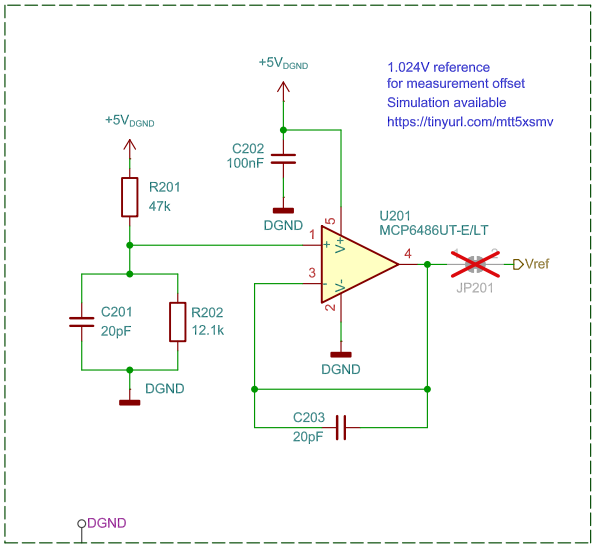


Inverter AC current measurement

VOUT=((RB/RC)VOUTP−(RA/RD)VOUTN)+VREF

If RB=RC and RA=RD
VOUT=G(VOUTP−VOUTN)+VRef
With VRef = 1V
G= 12.1k/10k = 1.21

VOUT = 1.21(VINP-VINN)+1V
FS = +-1.01Vpeak for differential input of +-0.835Vpk
Output centered at 1V. Full scale 0-2V



NOTES :

Source repository : https://github.com/owntech-foundation/uVerter

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	A				
2026-01-27	-	JAL	JAL	JAL	

Grid current measurement

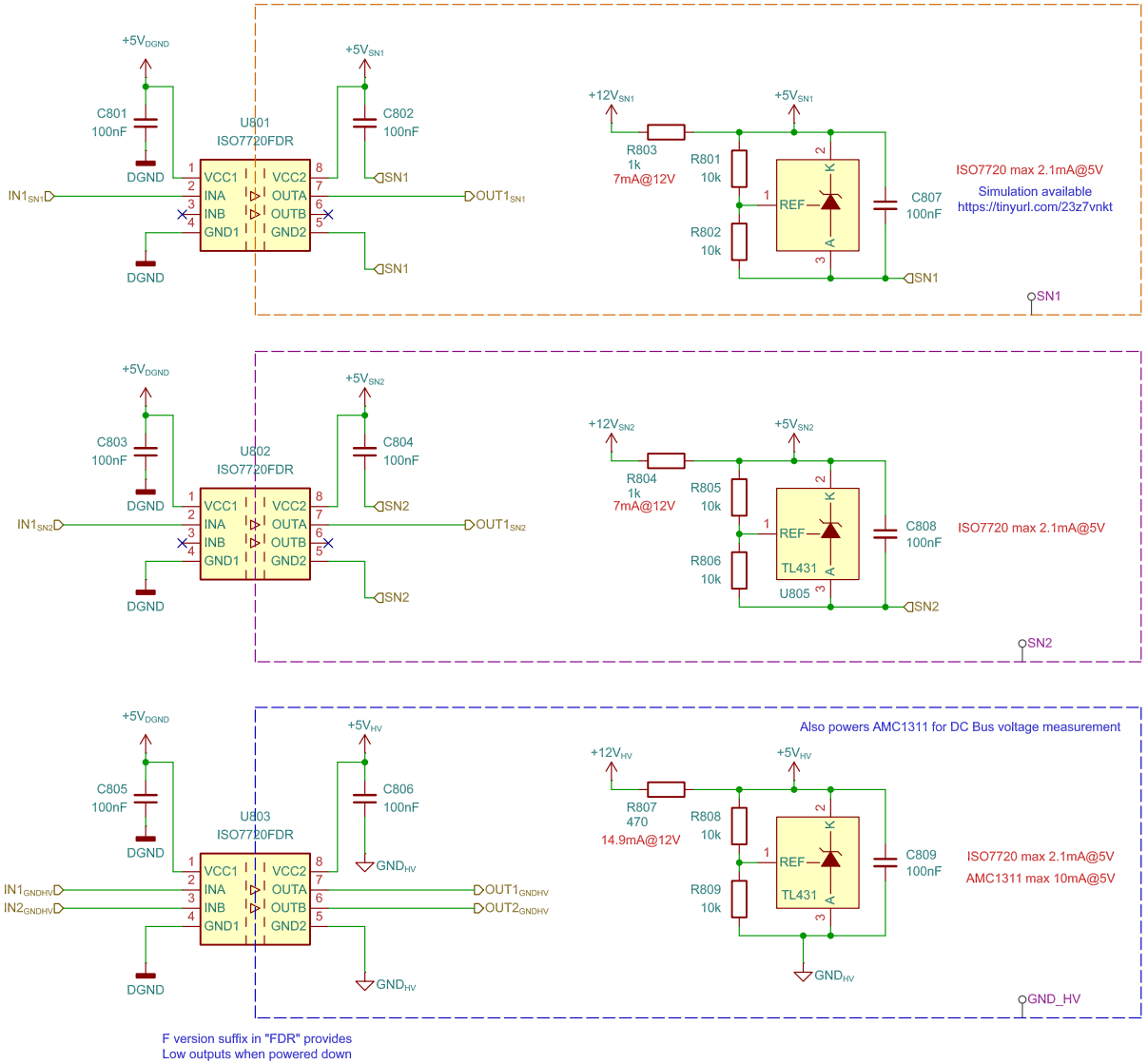
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File: current.kicad_sch

Size: A4 | Sheet: /Power stage/AC Current sense/

KiCad E.D.A. 10.0.5 | Id: 6/9



NOTES :

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Inverter PWM isolators

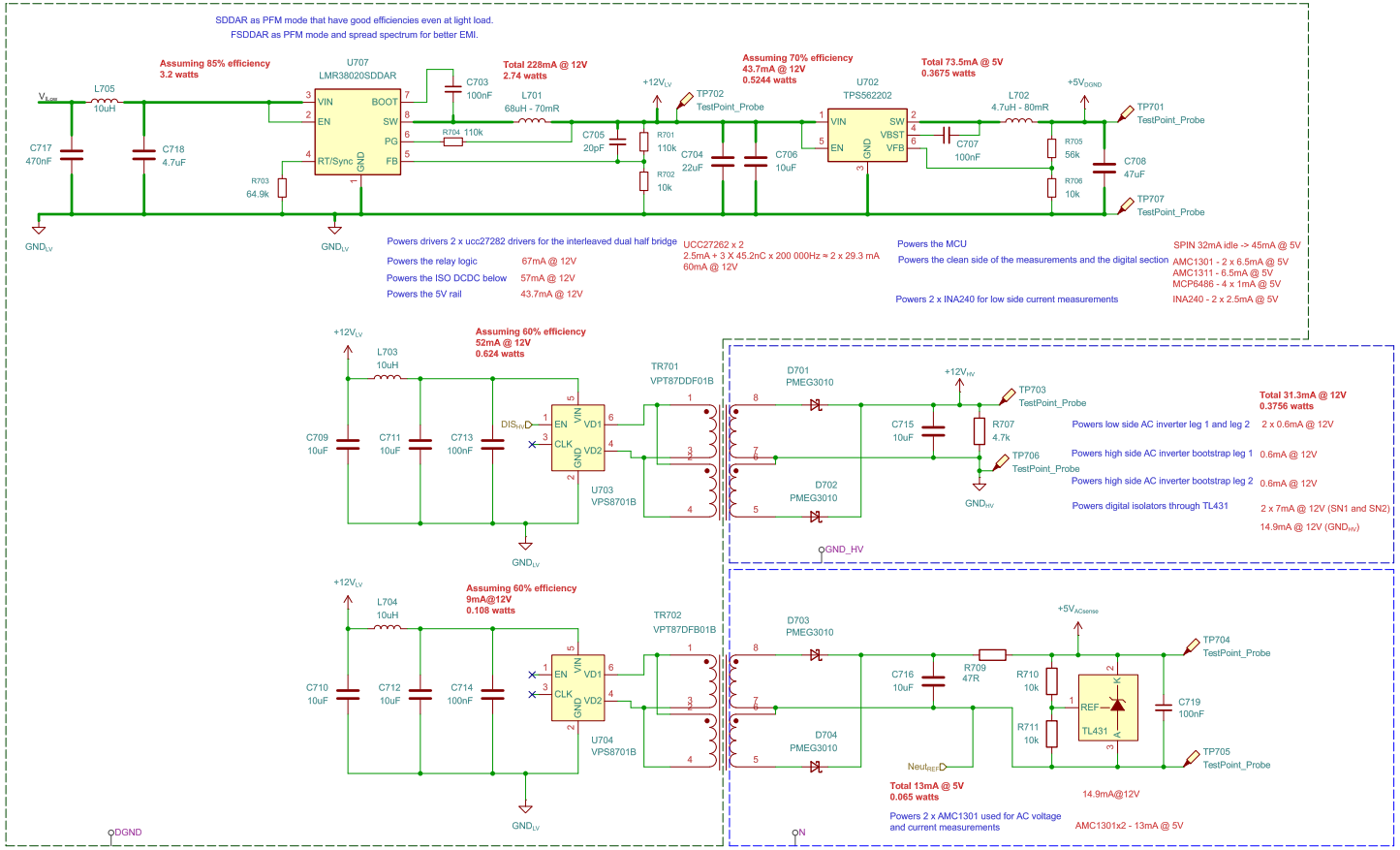
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File: signal_isolators.kicad_sch

Size: A4 | Sheet: /PWM Signal Isolators/

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NOTES :

Source repository : https://github.com/owntech-foundation/uVerter

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2026-01-27	-	JAL	JAL	JAL	

Internal power supplies

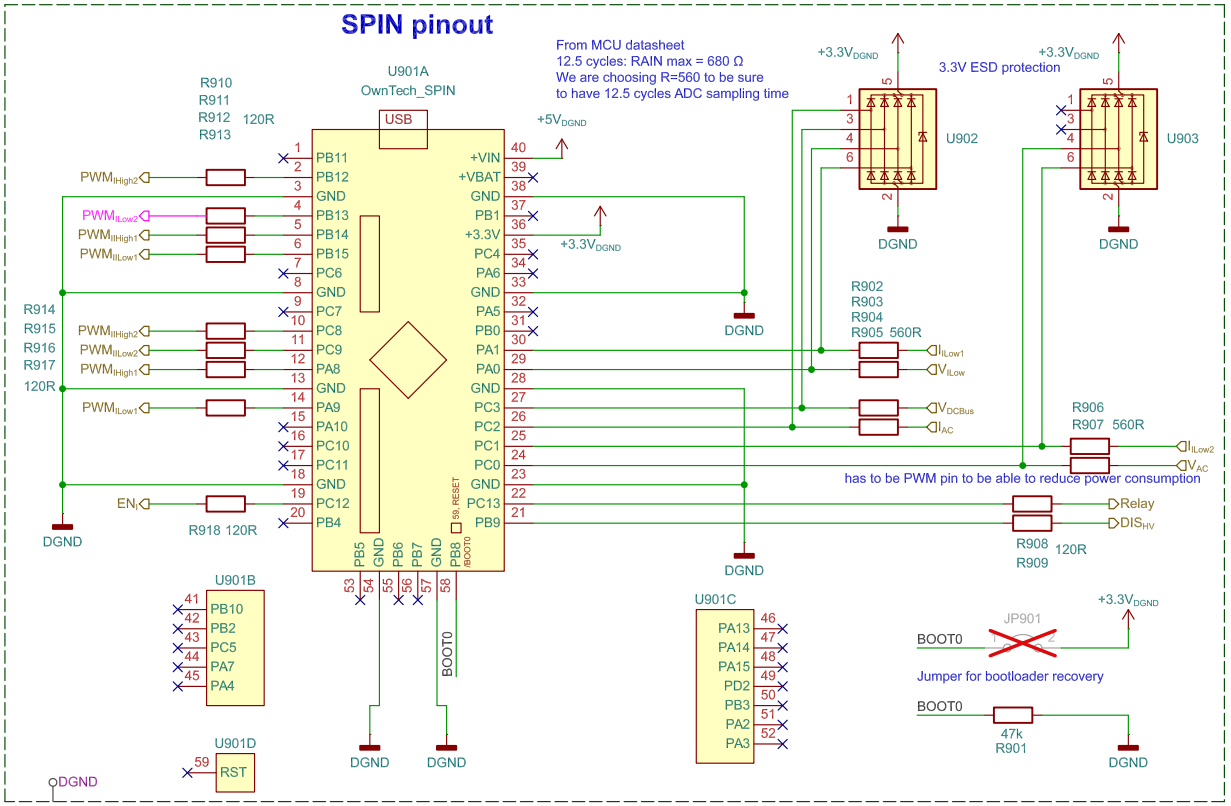
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File: feeder.kicad_sch

Size: A3 | Sheet: /Internal power supplies/

KiCad E.D.A. 10.0.5 | Id: 8/9



Sensors correspondance with TWIST board :

µVerter	TWIST
I_{ILow1}	I_{ILow1}
I_{ILow2}	I_{ILow2}
V_{ILow}	V_{ILow1}
V_{DCBus}	V_{IHigh}
I_{AC}	I_{IHigh}
V_{AC}	V_{ILow2}

µVerter	PWM Channel
PWM_{IHigh1}	CHA1
PWM_{ILow1}	CHA2
PWM_{IHigh2}	CHC1
PWM_{ILow2}	CHC2
$PWM_{IIHigh1}$	CHD1
PWM_{IILow1}	CHD2
PWM_{IILow2}	CHE1
PWM_{IILow1}	CHE2
Relay	TIM1 CH1n

NOTES :

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Power stage controller

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File: Microcontroller.kicad_sch

Size: A4 | Sheet: /Microcontroller/

KiCad E.D.A. 10.0.5 | Id: 9/9